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Teacher-guided AI-assisted development of a virtual single-slit diffraction laboratory for Grade 10 physics: a classroom case study

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Abstract

This classroom case study describes the teacher-guided AI-assisted development of a virtual single-slit diffraction laboratory for Grade 10 physics. A generative AI system was used as a rapid prototyping partner to create and iteratively refine an interactive simulation, on-screen measurement ruler, and supporting digital laboratory worksheet. The development process was shaped by repeated teacher feedback on classroom usability, conceptual accessibility, visual clarity, measurement precision, and error analysis. The final activity allowed students to vary wavelength, slit width, and screen distance, observe the resulting diffraction pattern, measure the central maximum, and compare measured values with the theoretical relation. Classroom implementation and exploratory student feedback are discussed in terms of engagement, visualization, usability, and perceived support for experimental understanding. The case illustrates how generative AI can support physics teachers in developing curriculum-linked virtual laboratory activities when the teacher remains central in directing the pedagogy, checking the physics, and adapting the tool for learners.

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Funding

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Author contributions

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References